

[§4 and §9 - adapted] For each statement, state if it is **True** or **False**.

- If it is true, give a direct proof.
- If false, provide a counterexample.

Problem 1. Suppose $a, b, c \in \mathbb{Z}$, if $a \mid bc$, then $a \mid b$ or $a \mid c$.

Problem 2. If two integers have the same parity, then their sum is even.

Problem 3. For every natural number n , the integer $n^2 + n + 41$ is prime.

Problem 4. If x is an odd integer, then x^3 is odd.

Problem 5. If A and B are finite sets, then $|A \cup B| = |A| + |B|$.

Problem 6. Suppose a is an integer. If $5 \mid 2a$ then $5 \mid a$.

Problem 7. Suppose A, B and C are sets. If $A \times C = B \times C$, then $B = C$.

Problem 8. Suppose x and y are positive numbers. If $x < y$, then $x^2 < y^2$.